EE2004: Digital Signal Processing

Tutorial 8

(March 26, 2026)

1. Consider a first order, causal, and stable allpass filter of the form

$$A_1(z) = \frac{1 - c_1^* z}{z - c_1}$$

Determine the expression for $1 - |A_1(z)|^2$ and then show that

$$1 - |A_1(z)|^2 \begin{cases} < 0 & \text{for } |z|^2 < 1 \\ = 0 & \text{for } |z|^2 = 1 \\ > 0 & \text{for } |z|^2 > 1 \end{cases}$$

From this deduce that the above property holds for an N-th order allpass filter.

2. In the previous problem, how do the inequalities in the expression for $1 - |A_1(z)|^2$ change if

$$A_1(z) = \frac{-c_1^* + z^{-1}}{1 - c_1 z^{-1}}$$

3. Consider the causal sequence $x[n]$ with z-transform

$$X(z) = \frac{\left(1 - \frac{1}{2} z^{-1}\right)\left(1 - \frac{1}{4} z^{-1}\right)\left(1 - \frac{1}{5} z\right)}{\left(1 - \frac{1}{6} z\right)}$$

For what values of a is $a^n\, x[n]$ a real-valued, minimum-phase sequence, i.e., has all its poles and zeros inside the unit circle ?

4. Eight finite-duration sequences are given below, all of which have identical magnitude frequency response (to within a scale factor). Which of the sequences has all of its zeros *inside* the unit circle?

(a) $\{20, -61, 46, -8\}$
(b) $\{32, -112, 106, -20\}$
(c) $\{-4, 29, -62, 40\}$
(d) $\{15, -102, 192, -96\}$
(e) $\{8, -46, 61, -20\}$
(f) $\{-32, 64, -34, 5\}$
(g) $\{10, -53, 56, -16\}$
(h) $\{200, -310, 145, -20\}$

5. Compute the "partial energy" of the sequences using the formula $E_k = \sum_{n=0}^{k} |h[n]|^2$ for $k = 0, \ldots, 3$ and normalize it by the sequences "total energy", viz., E_3. Can you identifiy the minimum phase sequences from partial energy sequences? From this it should be clear that "minimum energy delay sequence" is a more apt term than "minimum phase sequence". Where are the zeros of the sequence that has the maximum energy delay ?

6. **Computer assignment** Consider the sequences given in Question 4. Plot the magnitude frequency responses of each sequence on a log-magnitude scale. If they have the same shape but are shifted versions of one another, find out the scale factors needed to make all the magnitude responses identical.

7. **Computer assignment** An FIR filter using the Parks-McClellan algorithm can be designed using the command `h = firpm(32,[0 0.3 0.4 1],[1 1 0 0])`. Let `H = freqz(h, 1, 4000)`. Plot magnitude in both linear and log scales and observe the differences. Plot the unwrapped phase angle and observe its properties. Plot `zplane (roots(h))` and verify that jumps of π occur in the phase response at unit circle zeros locations. Let $x[n] = \cos(\omega_1 n)$, where $\omega_1/2\pi = 0.1$. Filter this sequence through the above filter by convolving the input with the impulse response. By suitably delaying the input are you able to align it exactly with the output? Ignore the transients at the beginning and at the end. Is this delay related to the length of the filter? Is it related to the slope of the phase response? Now reduce the length of the filter by one by changing the first argument of `firpm`. Are you now able to align exactly the input and the filtered output? What is the slope of the phase response? Explain your answer.

8. **Computer assignment** Let us compare how an IIR filter with almost identical magnitude response, but whose phase is highly nonlinear, behaves when an input with two tones is applied to it. An *elliptic filter* can be designed using the command `[b,a] = ellip(5,0.3,35,0.3)`. Compare the magnitude response of this filter with the one designed in the previous example. Let us consider the response to $x[n] = \cos(\omega_1 n) + \cos(\omega_2 n)$, where ω_1 is as before and $\omega_2/2\pi = 0.15$. Note that the second component is at the band edge, where the magnitude is still close to unity for both filters, but the phase response changes rapidly for the elliptic filter. Plot the group delay using `grpdelay(b, a, 4000)`. How different are the values for the two tones? Compute the output using `filter(b, a, x)` and plot the result. Is it possible to align the elliptic filter's input and output by suitably delaying one of them? Carefully contrast this filter's behaviour (nonlinear phase response) with that of its counterpart in the previous problem (exactly linear phase response). Add a third tone with frequency $\omega_3/2\pi = 0.4$ to the input and verify that both filters attenuate it significantly, illustrating frequency selective filtering. You can also add noise (use the command `randn`) and see how the components that fall in the stopband get attenuated.

9. **Computer assignment** Compute and plot the phase response using the functions `freqz`, `angle`, `phasez`, `unwrap`, and `phasedelay` for the following systems:
   (a) The pure delay system given by $y[n] = x[n-15]$
   (b) $H(z) = \dfrac{1 + 1.655\,z^{-1} + 1.655\,z^{-2} + z^{-3}}{1 - 1.57\,z^{-1} + 1.264\,z^{-2} - 0.4\,z^{-3}}$.

10. **Computer assignment** Compute the group delay of the system

$$H(z) = \frac{1 + z^{-1} + z^{-2} + z^{-3}}{1 + 0.9\,z^{-1} + 0.81\,z^{-2} + 0.927\,z^{-3}}$$

using the MATLAB function `grpdelay`.